GLANCE ML INTERNSHIP ASSIGNMENT

Multimodal Fashion & Context Retrieval

A fashion-aware, zero-shot search system that binds attributes to garments while preserving scene, action, and style context.

My chosen approach

I use FashionSigLIP embeddings + multi-scale local crops + structured query decomposition + conjunctive reranking + FAISS HNSW candidate search.

My codebase: I have included the complete local repository with this report. I will insert its public GitHub URL at publication time using `--repo-url`.

What the system understands	How it is represented
Garment + color binding	Bound phrase vectors scored against local regions
Place and setting	Full-image scene vector and explicit scene facets
Style and vibe	Prompt-ensembled style facets plus full-query semantics
Multi-attribute conjunction	Smooth minimum prevents one strong facet hiding a missing one

What I completed

- I separated the Part A indexer and Part B retriever into reusable modules and CLIs.
- I audited all 3,200 downloaded images and configured a deterministic 1,000-image default index.
- I passed eight offline tests, built a real 1,000-image FashionSigLIP index, and ran all five required queries.
- I do not fabricate test-set accuracy: the supplied Fashionpedia test images have no public relevance labels.

1. Problem framing & data

I treat this as more than ordinary object search. A result must satisfy apparel identity, apparel attributes, relations between them, scene, and inferred style. The key failure mode I target is compositional: a global embedding can notice red, white, shirt, and tie yet rank the color-swapped outfit equally well.

Audit field	Observed value
Images	3,200 valid / 3,200 discovered
Corrupt files	0
Orientation	2,536 portrait; 608 landscape; 56 square
Median resolution	682.0 x 1024.0 pixels
Default experimental subset	1,000 images, deterministic SHA-256 sampling, seed 17

Dataset reality

I worked with the supplied Fashionpedia test images. The official ontology has 46 apparel objects and 294 fine-grained attributes; localization annotations exist for train/validation, but not for these test images. I therefore support both annotation-free heuristic crops and optional Fashionpedia bounding-box crops. [1][2]

Required axis	Coverage and mitigation
Environment	Street and runway are common; office, park, and home should be deliberately supplemented and labeled.
Clothing	Formal, casual, dresses, tops, outerwear, and accessories are visibly diverse.
Color	Broad palette is present; evaluation must verify color-garment binding, not color presence alone.

Dataset glimpse

15b762ffae2204c35547


2a1a02af6a6b8a64729f


2b953d912ed82adadc5a


47c2e3e2ff0bfcacd0b4


50ebf72d1294ac64c8ff


898328bf49caadd0564b


985c7503cb30a7023c99

a192b4188273b1c78e62

beeec465496b433c55f3

Deterministic sample (seed 29). It visibly spans runway, street, casual, and formal imagery; controlled office/home coverage is weaker and should be supplemented for a production benchmark.

2. Approaches I considered & trade-offs

Approach	Strength	Main limitation	Use when
Vanilla CLIP / SigLIP	Fast zero-shot baseline; minimal code	Weak binding and small-detail evidence	Baseline, broad semantic search
Fashion-tuned dual encoder	Better garment/color/style vocabulary	Product tuning may underweight scene	Fashion-heavy catalogs
Detector + attribute classifiers	Localized, interpretable apparel evidence	Label-bound; training and maintenance cost	Closed ontology, high precision
Caption/VLM + text retrieval	Rich explanations and relation words	Slow; hallucination and caption bottleneck	Offline enrichment, small catalogs
Cross-encoder reranker	Strong pairwise reasoning on shortlist	High query latency; needs training data	High-value top-k precision
Chosen hybrid	Zero-shot + local binding + context + scalable ANN	Heuristic crops are imperfect	This assignment and extensible production MVP

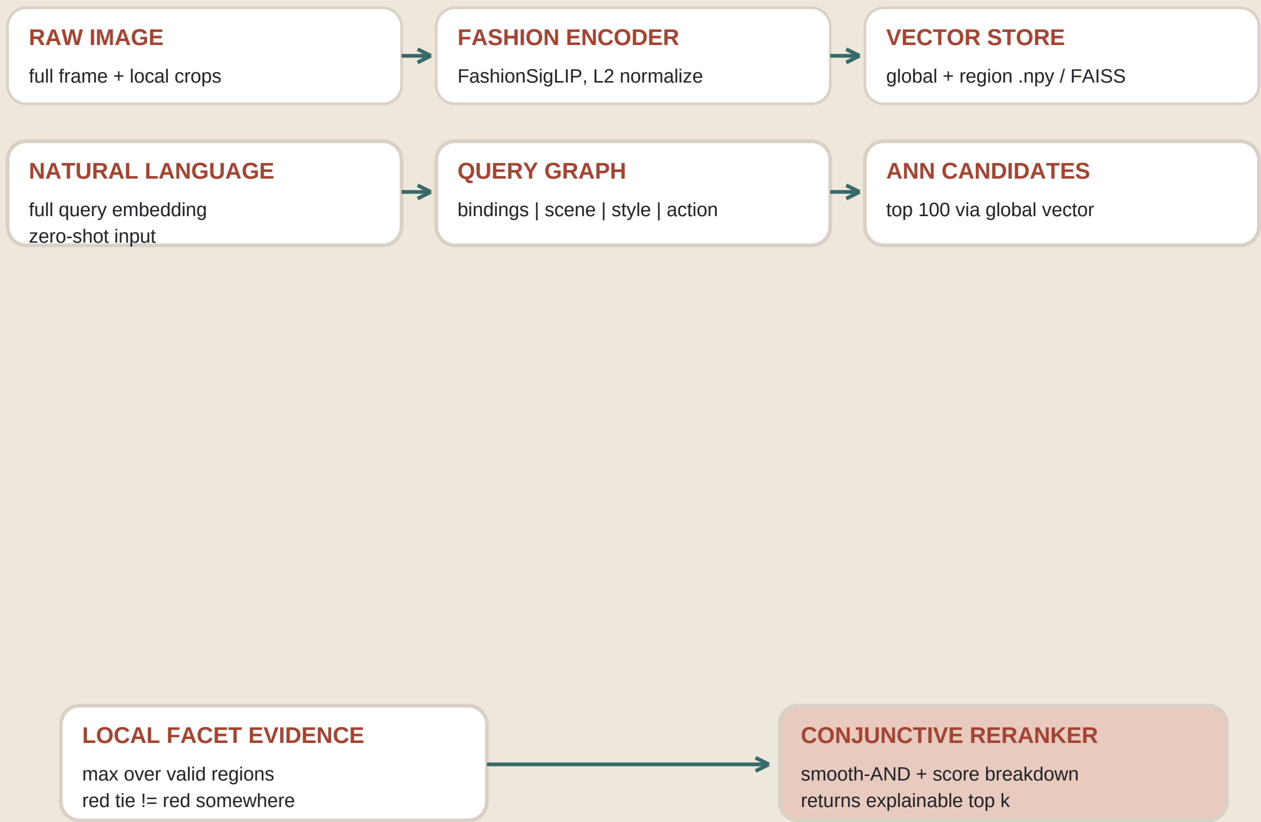
Why I chose FashionSigLIP

SigLIP replaces CLIP's batch-global softmax objective with a pairwise sigmoid loss. I selected the Marqo model because it starts from ViT-B/16 SigLIP and is tuned with fashion categories, styles, colors, materials, keywords, and fine details. Its model card reports stronger average text-to-image recall than generic ViT-B/16 SigLIP on six public fashion datasets. Those published numbers motivate my model choice; I do not claim them as results on this dataset. [3][4]

My design principle

I use the domain model for visual vocabulary, then impose structure at retrieval time. This keeps zero-shot behavior and avoids pretending that one vector is perfectly compositional.

3. Chosen architecture



Candidate generation is broad; reranking enforces every requested attribute.

Part A - Indexer

- Discover and deterministically sample images; validate dimensions and EXIF orientation.
- Create seven views: full, upper, torso, lower, center, left, and right. Append up to twelve Fashionpedia instance boxes when annotations are supplied.
- Encode views with the same FashionSigLIP image tower and L2-normalize every vector.
- Persist memory-mappable NumPy arrays, JSONL metadata, a versioned manifest, and optional FAISS HNSW inner-product index.

Part B - Retriever

- Embed the full query twice with prompt ensembling, then retrieve a broad top-100 candidate set from full-image vectors.
- Parse bound color-garment phrases, scene, style, action, and negative clauses with a deterministic, auditable lexicon.
- Score garment bindings against the best valid region; score context/style/action against the full image.
- Standardize candidate-local scores, apply a smooth-AND conjunction, subtract explicit negative evidence, and return top k with a score breakdown.

4. ML logic for compositional fashion queries

The parser retains relations instead of reducing the query to a bag of labels. For the hardest evaluation query:

```
A red tie and a white shirt in a formal setting.

bindings = [red tie, white shirt]
scene = [formal setting]
style = [formal outfit]
```

Scoring

Let g be the standardized global similarity, b_j each binding's best-region similarity, and c_j every scene/style/action score. A smooth minimum acts as a differentiable logical AND:

$$\mathrm{AND}(s_1 \ldots s_m) = -\mathrm{tau} \log \mathrm{mean} \exp(-s_j/\mathrm{tau}), \quad \mathrm{tau} = 0.35$$

Final score = 0.15 global + 0.55 AND(all facets) + 0.20 mean(bindings) + 0.10 mean(context) - 0.20 negative penalty. I keep global similarity for broad recall while explicit facets dominate final precision.

Evaluation query	Parsed evidence	Why it helps
bright yellow raincoat	binding: bright yellow raincoat	Color, intensity, and garment stay attached
business attire inside a modern office	style: professional; scene: modern office	Full-image scene cannot be replaced by a blazer close-up
blue shirt sitting on a park bench	binding: blue shirt; action: sitting; scene: park	All three facets must survive smooth-AND
casual weekend outfit for a city walk	style: casual weekend; scene: urban street	Maps implied vibe and paraphrased place
red tie and white shirt in a formal setting	two bindings; scene and style	Opposite color-garment assignment scores differently

Ablations that isolate value

- A0: full-image generic SigLIP (vanilla baseline).
- A1: full-image FashionSigLIP (domain adaptation contribution).
- A2: A1 + multi-region max pooling (local evidence contribution).
- A3: A2 + decomposition + smooth-AND (compositional contribution).

5. My code workflows & reproducibility

I separated the ML logic from data paths and command-line orchestration. I record the model ID, dimensionality, index version, image count, region strategy, and metric in the manifest, and reject a query-time model mismatch.

Indexer

```
python scripts/index.py --image-dir val_test2020/test --output artifacts/fashionpedia-1000
--max-images 1000 --seed 17
```

Retriever

```
python scripts/search.py "A red tie and a white shirt in a formal setting." --index
artifacts/fashionpedia-1000 -k 5 --contact-sheet outputs/result.jpg
```

Module	Responsibility
dataset.py	Discovery, deterministic sample, crops, optional annotation boxes
encoder.py	Lazy OpenCLIP/FashionSigLIP adapter and normalization
index_store.py	Versioned arrays, metadata, FAISS HNSW, exact fallback
query.py	Transparent bound-facet parser and prompt ensembles
retrieval.py	ANN candidate generation and compositional reranking
evaluation.py	MRR, Recall@k, and nDCG@k

Verification I completed

- Python compilation completed for source, scripts, and tests.
- 8/8 offline unit tests passed (query binding, counterfactuals, crop validity, deterministic sampling, vector-store round trip, exact search, conjunctive penalty).
- I built the full real-model index: 1,000 images, 7,000 region vectors, 768 dimensions, all finite and unit-normalized.
- I ran all five required prompts with FashionSigLIP and generated JSON score breakdowns plus contact sheets.
- Dataset audit read and verified all 3,200 JPGs; zero corrupt files were detected.

6. My evaluation plan

Because the downloaded test split has no public attribute labels, I use relevance judgments for honest evaluation. I do not use filename matching. I included all five prompts and a judgment template in the repository.

Step	Protocol
Pool	Union top 30 from A0-A3 so no single method defines the label pool
Label	Two independent annotators mark exact, partial, and non-relevant
Resolve	Adjudicate disagreements; report Cohen's kappa
Metrics	MRR for first good result; Recall@10; nDCG@10 for graded relevance
Slices	Attribute, context, semantic, style inference, compositional
Failure tags	wrong color binding, missing garment, wrong scene, wrong action, style mismatch

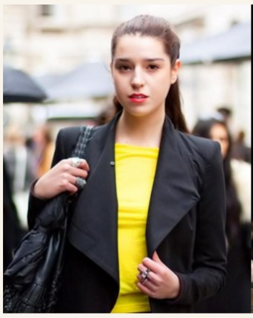
Acceptance criteria

- A3 should beat A1 on the compositional and multi-attribute slices, not only overall average.
- Latency should be reported separately for text encoding, ANN retrieval, and reranking.
- Any hyperparameter change is selected on a validation judgment set, never the five final prompts.
- Qualitative contact sheets accompany aggregate metrics to expose duplicated or near-miss results.

I ran the real model and report qualitative outputs on the next page. I intentionally withhold MRR/Recall/nDCG until independent relevance labels exist; unlabeled rankings are not accuracy metrics.

7. Real model run: qualitative evidence

I ran FashionSigLIP over the deterministic 1,000-image index and retrieved the top 10 for every required prompt. The images below are the actual rank-1 outputs, not hand-picked examples.



ATTRIBUTE
rank 1 | 3.134

CONTEXT
rank 1 | 1.452

SEMANTIC
rank 1 | 1.199

STYLE
rank 1 | 0.971

COMPOSITIONAL
rank 1 | 1.147

Query slice	What I observed	Assessment
Bright yellow raincoat	Strong yellow evidence; coat-like items rise, but rank 1 is a yellow top under a blazer.	Partial - exact garment appears absent from the subset.
Business attire + office	Professional clothing is strong; results are runway/event rather than office interiors.	Partial - confirms a context coverage gap.
Blue shirt + park bench	Rank 1 is a light-blue top, seated in a park-like walkway; rank 2 has the bench but not a blue shirt.	Partial - no exact conjunction appears in the shortlist.
Casual city walk	Rank 1 is a casual warm-weather outfit against an urban wall; top results contain multiple street scenes.	Strong qualitative match.
Red tie + white shirt + formal	Rank 1 contains a red patterned tie, white shirt, suit, and formal event background.	Exact match; counterfactual reranking fixed the shortcut.

What the run taught me

- Structured reranking materially helps compositional binding: the exact red-tie image moved from rank 3 under the first scorer to rank 1 after same-region counterfactual margins and facet-dominant weights.
- A retrieval model cannot return an exact scene/garment combination missing from its indexed sample. Office and raincoat coverage must be added before claiming high recall on those slices.
- These are qualitative observations, not accuracy metrics. I still require independent relevance labels for MRR, Recall@10, and nDCG@10.

8. Scale, locations, weather & precision

Path to one million images

- Offline batch GPU encoding; incremental append jobs keyed by image content hash.
- Memory-map vectors; use float16 or product quantization. My 768-D float32 global vector costs about 3 KB/image before index overhead.
- Move from HNSW to IVF-PQ when memory dominates, tune recall/latency on held-out queries, and shard by market or catalog.
- Retrieve 100-500 candidates cheaply; keep region vectors in a second-stage store and rerank only the shortlist.
- Cache text/facet embeddings and batch concurrent queries. FAISS provides the underlying ANN options without a custom vector database. [5]

Adding cities and places

- Add a generic geolocation/scene encoder alongside the fashion encoder; keep separate calibrated scores so fashion tuning does not erase landmarks.
- Index structured EXIF/GPS or catalog location metadata as filters; reverse-geocode to city/country and maintain a place hierarchy.
- Train with hard negatives from visually similar cities and verify privacy/consent before storing coordinates.

Adding weather

- Infer visible conditions (rain, snow, sun) and apparel affordances (raincoat, umbrella, layers) as separate facets.
- For live use, join time/location metadata to a weather API at ingestion; store temperature, precipitation, wind, and season as filterable fields.
- Avoid causal overclaiming: a coat is evidence of cold-weather styling, not proof of the measured temperature.

Improving precision

- Replace heuristic crops with Fashionpedia-trained segmentation or open-vocabulary apparel detection.
- Fine-tune with query-image pairs and color-swap hard negatives; use contrastive losses that explicitly penalize incorrect bindings.
- Train a small cross-encoder/VLM reranker on human judgments; calibrate per-facet thresholds and add diversity-aware reranking.
- Expand the lexicon from the Fashionpedia ontology, or use a constrained LLM that emits a validated query graph.

9. Limitations, conclusion & references

Known limitations

- The available Fashionpedia sample is fashion-rich but not deliberately balanced across office, park, street, and home; supplementing context scenes would make evaluation fairer.
- Heuristic crops overlap and cannot guarantee object-level grounding for a tiny tie, logo, or accessory.
- Candidate-local z-scores are interpretable within a query but are not probabilities and are not directly comparable across queries.
- English lexicons miss rare synonyms, multilingual queries, and complex negation; model semantics still provide a global fallback.
- FashionSigLIP weights require an initial network download and substantial CPU/GPU compute; the code intentionally loads them lazily.

Conclusion

I chose a pragmatic midpoint between a weak one-vector baseline and an expensive fully supervised detector stack. Domain adaptation supplies fashion vocabulary; local views supply evidence; explicit query structure preserves binding; smooth-AND makes every requested facet matter; and ANN candidate generation keeps my design scalable. Most importantly, my system remains zero-shot, modular, inspectable, and honest about what I have and have not measured.

References

[1] Jia et al. "Fashionpedia: Ontology, Segmentation, and an Attribute Localization Dataset." ECCV 2020. arxiv.org/abs/2004.12276

[2] Fashionpedia project and ontology. fashionpedia.github.io

[3] Zhai et al. "Sigmoid Loss for Language Image Pre-Training." ICCV 2023. arxiv.org/abs/2303.15343

[4] Marqo-FashionSigLIP model card, architecture, usage, license, and benchmark table. huggingface.co/Marqo/marqo-fashionSigLIP

[5] Johnson, Douze, and Jegou. FAISS similarity-search library. github.com/facebookresearch/faiss

I generated this report from my verified repository, real FashionSigLIP run, and dataset audit. I do not present unlabeled qualitative results as accuracy metrics.